

	ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		Assignment
Branch	CSE/CSIT	Programme	B.Tech
Course Name	Introduction to the Theory of Computation	Semester	5th
Course Code	CSE3731	Academic Year	2025-26
ASSIGNMENT - 2			
Submission due date: 23/12/2025			
Learning Level (LL)	L1: Remembering	L3: Applying	L5: Evaluating
	L2: Understanding	L4: Analysing	L6: Creating
Q's	Questions		COs
1	a. Consider a grammar G is given as follows: 1. $S \rightarrow AB \mid aaB$ 2. $A \rightarrow a \mid Aa$ 3. $B \rightarrow b$ Determine whether the grammar G is ambiguous or not. If G is ambiguous, construct a leftmost and a rightmost derivation for the string "aab"? b. Construct a CFG for a language $L = \{ \text{The total number of 'a' in a word } w \text{ is even} \mid w \in \{a, b\}^* \}$		CO4
2	a) Construct a CFG for a language $L = \{a^n b^{2n} c^m \mid n, m \geq 1\}$ b) Consider the following production: $S \rightarrow aA \mid bB$ $A \rightarrow aA \mid \epsilon$ $B \rightarrow bB \mid \epsilon$ $C \rightarrow cD$ $D \rightarrow d$ What are the useful productions after removing all the useless symbols from the above grammar?		CO4
3	a) Explain the Chomsky Hierarchy and show the relationship between the different types of grammars. b) Construct a grammar in Chomsky normal form(CNF) from the equivalent CFG, G, with production rules $S \rightarrow aAbB$, $A \rightarrow aA \mid a$, $B \rightarrow bB \mid b$.		CO4
4	a) Prove that the language $L = \{0^n \mid n \text{ is a prime}\}$ is not a Context-Free Language. b) Prove that the language $L = \{ww \mid w \in \{0,1\}^*\}$ is not a Context-Free Language using the pumping lemma.		CO4
5	a) Design a PDA to accept the language $L = \{a^n b^n \mid n \geq 0\}$. Where the stack symbol $\sigma = \{x, \\$\}$. Let the stack symbol x represent counting the number of string symbols 'a' for the stack, along with the initial stack symbol is \$. b) Convert the CFG $S \rightarrow 0S1 \mid \epsilon$ to an equivalent PDA		CO4
6	a) Construct the PDA that recognizes the Language $L = \{ww^R \mid w \in \{0,1\}^*\}$. Show the stack implementation for the string 10100101. b) Design a PDA that recognizes the Language $L = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i=j \text{ or } i=k\}$. Show the stack implementation for the string aaabbccc.		CO4

7	a) Construct a TM that decides the language $L = \{w\#w \mid w \in \{0,1\}^*\}$ b) Construct a Turing Machine that recognizes the language $L = \{0^{2^n} \mid n \geq 0\}$	CO5	L3,L5
8	a) Given a Turing machine M over the alphabet $\{0,1\}$, is it decidable to determine whether M accepts exactly the set of all strings that contain an even number of 0s? b) What is the difference between a recursive language and a recursively enumerable language?	CO5	L3,L5
9	a) Given two Turing machines, M_1 and M_2 , is the problem of deciding if $L(M_1) = L(M_2)$ decidable? b) What are P, NP, NP-Complete, and NP-Hard problems? Explain the relationship of these concepts with the help of a Venn diagram and give some examples for each problem.	CO6	L1, L2
10	a) Show that if $P=NP$, then every language $A \in P$, except $A = \Phi$ and $A = \Sigma^*$ is NP-Complete. b) Proof that if $HALT_{TM} = \{ \langle M, w \rangle \mid M \text{ is a Turing Machine and } M \text{ halts on input } w \}$ then $HALT_{TM}$ is undecidable.	CO6	L2,L4

Course Outcomes	By the end of the course, through lectures, readings, home works, assignments, and exams, students will be able to:	
	CO1	Acquire knowledge of automata theory and develop proficiency in using mathematical terminology and proof strategies relevant to computational theory
	CO2	Design deterministic and non-deterministic finite automata for regular languages
	CO3	Develop and analyse regular expression for regular languages, and establish their equivalence with finite automata through bidirectional conversion.
	CO4	Design context free grammars and corresponding push down automata for the recognition of context free languages.
	CO5	Design and analyse Turing machines for solving computational problem
	CO6	Enhance proficiency in analysing computational problems concerning decidability, undecidability, complexity classifications and reducibility principles.

- ✓ Assignment scores/markings depend on neatness and clarity.
- ✓ Plagiarized assignments will be given a zero mark.
- ✓ Submit the hard copy of your assignment by the due date, i.e. **23.12.2025**
- ✓ Submit the assignment handwritten on A4 size papers and spirally bound to your ITC class teacher. A front page must be present containing the details of the subject, the assignment and the student.